1 Extension: Addressing Class Imbalance

Our extension introduces three imbalance-focused mechanisms that help our model better capture the inherently skewed label distribution of the toxic comment classification task.

- **Class-Adaptive Focal Loss (α and γ).** We replace the baseline's fixed scalar α with a per-class α vector derived from inverse label frequencies and clipped by $\alpha_{\max}$. We also tune the focusing parameter γ.
  - Class-specific α increases the contribution of rare labels. Clipping prevents extremely rare classes from causing unstable gradients.
  - We varied the focusing parameter to control how much the model down-weights easy examples. When $\gamma = 0$, the focal loss reduces to *weighted BCE*.
- **Bias Initialization:** We tested the impact of initializing the final layer's bias to $b = -\log((1-\pi)/\pi)$ (where π is the prior probability) versus standard initialization. This was done to see if explicitly aligning the initial output with the data distribution accelerates convergence or if the model learns better from a neutral starting point.
- **Dynamic Per-Class Threshold Tuning.** Rare classes rarely cross a 0.5 probability, even when predicted correctly. We implemented a dynamic thresholding mechanism that analyzes the Precision-Recall curves of the training data to calculate the optimal decision boundary for each specific class.

We also switch to Macro-F1 as our primary evaluation metric instead of AUC-ROC, since AUC was already near-saturated in earlier experiments and no longer differentiated model performance. Macro-F1 allows us to directly evaluate precision and recall for each class, providing a much clearer picture of model behavior under severe class imbalance.

1.1 Empirical Evaluation

We conduct hyperparameter tuning on $\alpha_{max} \in [0.6, 0.75, 0.9]$, $\gamma \in [0.0, 2.0, 4.0]$, and $init\_bias \in [True, False]$ to determine the settings that most effectively address class imbalance, evaluated on the development set. We attached the tuning log in the submission for reference. We then assessed our extension on the test set using the optimal hyperparameters identified during tuning ($\alpha_{max} = 0.6, \gamma = 2.0, init\_bias = False$). Table 1 summarizes the results comparing with the strong baseline compared to the previous milestone.

Table 1: Performance comparison on the test set.

Metric	Strong Baseline	Extension
Mean AUC-ROC	0.9924	**0.9931**
Macro-F1	0.6751	**0.6883**
Macro-Precision	**0.7439**	0.6601
Macro-Recall	0.6260	**0.7264**

The results demonstrate a successful rebalancing of the precision-recall trade-off. The baseline achieved high precision but exhibited limited sensitivity to the minority classes. Our extension addressed this limitation and raised the primary **Macro-F1 score to 0.6883**.